

The Private Equity Advantage: Leveraged Buyout Firms and Relationship Banking

Victoria Ivashina

Harvard University and NBER

Anna Kovner

Federal Reserve Bank of New York

This article examines the impact of leveraged buyout firms' bank relationships on the terms of their syndicated loans. We examine a sample of 1,590 loans financing private equity sponsored leveraged buyouts between 1993 and 2005, and find that private equity firms' bank relationships are an important factor in cross-sectional variation in the loan interest rate and covenant structure. Our results indicate that bank relationships formed through repeated interactions reduce inefficiencies from information asymmetry and allow leveraged buyouts sponsored by private equity firms to occur on favorable loan terms. A one-standard-deviation increase in bank relationship strength is associated with an 8-basis-point (3%) decrease in the spread and a 0.21-basis-point (4%) increase in the maximum debt to EBITDA covenant. We also find evidence that banks price loans to cross-sell other fee business. A one-standard-deviation increase in both bank relationship strength and cross-selling potential translates into as much as a 4-percentage-point increase in equity return to the leveraged buyout firm. (*JEL* G20, G21, G23, G24, G32)

What is the role of leveraged buyout (LBO) firms? We propose and document financial intermediation as one source of value creation by LBO firms, in that these firms enable their portfolio companies to borrow on more favorable terms. We argue that an LBO firm's repeated interactions with the banks that finance its portfolio companies lower the costs of debt by reducing asymmetric information between the banks and the LBO firm (the financial sponsor).¹ The cost of financing is lower because the bank has acquired private

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¹ Throughout the text, we use the following terms interchangeably: LBO firm, private equity firm, and financial sponsor. A loan is "sponsored" by an LBO firm when the firm provides the equity capital that finances the LBO.

information about the LBO firm from prior transactions, which might, for example, enhance its confidence in the firm's due diligence process. We also find that banks offer better terms because they want to sell other fee-based services to LBO firms. This finding complements the prevailing view, which originated with Jensen (1986) and has been documented by Kaplan (1989), that LBOs create value by providing management with strong incentives to improve operating performance.²

The leveraged buyout industry has grown dramatically since the late 1980s. In the United States, LBO firms raised over \$148 billion in 2006 alone, and accounted for almost 17% of all mergers and acquisitions in that year. A substantial portion of these transactions were financed with syndicated bank loans.³ LBO firms are important clients for banks because of the frequency and scale of their transactions. In 2006, the top ten global LBO firms paid almost \$5 billion in investment banking advisory fees.⁴ Anecdotally, many LBO firms believe that their bank relationships provide them with a competitive advantage and that the banks with which they have stronger relationships are better able to "understand" more complicated transactions. However, the opposing view suggests that leveraged loans are commodities.

We analyze the impact of LBO firms' relationships with banks on loan terms (spread paid over LIBOR and financial covenants) using cross-sectional variation in the terms of syndicated loans sponsored by LBO firms. First, we hypothesize that if costly inefficiencies from information asymmetry are reduced when the same bank repeatedly lends to borrowers sponsored by the same LBO firm (strong bank relationship), then loans from this bank would be associated with better loan terms. Second, we test whether the prospect of additional business from LBO firms' investments, such as future underwriting or merger and acquisition fees, changes loan terms.

For a given loan, our primary measure of the strength of the bank relationship is the percentage of the amount of the loans sponsored by the LBO firm that were underwritten by the lead bank in the past five years. We proxy for future fee business with past fee business—the value of transactions sponsored by the LBO firm and advised by the lead bank in the past five years. After controlling for LBO firm size and the observable characteristics of the borrower and the loan, we find that a one-standard-deviation increase in an LBO firm's bank relationship decreases the loan spread on transactions it sponsors by 8 basis points and increases the maximum debt to EBITDA covenant by 0.21

² See Kaplan and Strömberg (2009) for a detailed overview of the literature.

³ Peak capital available for LBOs in 1987 was \$10.2 billion (*Wall Street Journal*, October 26, 2006). Between 2000 and 2005, loans comprised 44% of the capital structure of large LBOs (68% of debt) on average (Standard & Poor's Leveraged Commentary and Data 2006).

⁴ Source: DealLogic database. For example, Blackstone paid almost \$675 million, KKR paid \$660 million, and Bain Capital paid almost \$500 million. We assume that 10% of the fee is paid at deal announcement and 90% at deal closure.

points. A one-standard-deviation increase in the amount of an LBO firm's past fee business results in a 12 basis point lower spread and a 0.20 point higher maximum debt to EBITDA covenant. These results are not mutually exclusive. A simultaneous one-standard-deviation increase in the bank relationship *and* in potential fees from cross-selling results in a 17-basis-point reduction in the spread (a 5% decrease from the mean spread) and a 0.4-point-increase in the maximum debt to EBITDA covenant (a 7% increase).

An alternative explanation for our findings would be selection—perhaps LBO firms bring their best deals to the banks with which they have the strongest relationships. To refute this explanation, we use variation in existing LBO firms' lender concentration before a transaction to instrument for the selection of a given bank. The motivation for this instrument is the premise that if an LBO firm allows its bank relationships to become too concentrated, its bargaining power may decline and it may have to share the benefits of reduced costs of information asymmetries with its banks (Rajan 1992). The instrumental variable approach more than doubles the economic impact of repeated lending.

In addition to instrumenting for bank relationships, we confirm that stronger bank relationships and higher potential for cross-selling result in better loan terms when controlling for bank and LBO firm fixed effects. We directly control for borrower quality (portfolio company) using Compustat, and we do not find a relationship between our variables of interest and several ex-post measures of borrower quality.

We estimate that lower spreads result in annual savings of approximately \$0.55 million for each investment of an LBO firm that has a one standard deviation stronger bank relationship and one standard deviation stronger cross-selling prospects, which represents an 85-basis-point increase in the LBO firm's equity return. If the LBO firm takes advantage of the higher maximum debt to EBITDA covenant resulting from a one standard deviation stronger bank relationship and one standard deviation stronger cross-selling prospects to replace equity with debt, equity returns will be 3.50 percentage points higher.⁵ The total increase in equity returns of more than 4 percentage points (9 percentage points using the instrumental variables approach) is economically significant in the context of mean equity returns of 17% and estimations of risk-adjusted performance (alpha) for leveraged buyouts in general, which range from 2% to negative (see Groh and Gottschalg 2007; Kaplan and Schoar 2005; and Phalippou and Zollo 2006). Of course, some of the savings from lower debt costs may be shared with the LBO firm's targets.

These results have implications beyond a simple prediction of differences in loan terms among LBO firms. The findings serve as evidence of the role of LBO firms as financial intermediaries sponsoring buyout transactions. Because LBO firms have repeated relationships with banks, they can achieve more

⁵ See Section 3.2 for details on this calculation.

leverage on better terms than standalone borrowers. A company that wanted to take itself private would not have the same caliber of existing bank relationships (nor the potential for cross-selling) as an LBO firm. By showing how LBO firms receive differential access to financing terms, we demonstrate how these intermediaries make it possible to access the finance necessary for leveraged buyouts.

Studies of sources of value creation in LBOs highlight the importance of financing to private equity (PE) investment returns, but they do not examine cross-sectional determinants of financing. For example, [Guo, Hotchkiss, and Song \(2010\)](#) find that returns to capital are greater when a deal is financed with a higher proportion of bank loans, and [Kovner \(2010\)](#) finds that the leverage amount is an important source of value creation. Similarly, [Groh and Gottschalg \(2007\)](#) emphasize the importance of leverage when estimating a positive alpha for LBO firms.

In addition, several papers provide evidence that aggregate LBO volume is responsive to debt market conditions (including [Kaplan and Stein 1993](#); [Axelson, Jenkinson, Strömberg, and Weisbach 2007](#); and [Ljungqvist, Richardson, and Wolfenzon 2007](#)). We add to these results by providing a better understanding of cross-sectional patterns and the role of LBO firms in accessing credit markets. In this regard, [Demiroglu and James \(2010\)](#) find that more reputable LBO firms receive better financing terms, although they do not identify the source of differential terms.

This article complements other research on the role of private equity firms that documents the contribution of LBO firms and managers to operating performance (such as [Kaplan 1989](#); [Baker and Wruck 1989](#); [Acharya, Hahn, and Kehoe 2009](#); and [Kaplan, Klebanov, and Sorensen 2008](#)). Venture capital researchers, for example, have documented private equity firms' role in working with company management ([Bottanuzzi, Da Rin, and Hellman 2008](#); [Zarutskie Rebecca 2010](#)).

We also contribute to the literature on relationship banking by examining the special nature of the relationship between these shadow borrowers and banks. A vast amount of theoretical and empirical literature examines the role of banking relationships. This literature dates back to [James \(1987\)](#), and [Lummer and McConnell \(1989\)](#), who show that the market values of firms increase when they borrow repeatedly from the same banks. Other seminal work in this area includes [Petersen and Rajan \(1994\)](#) and [Berger and Udell \(1995\)](#), who find that banking relationships are associated with better financing terms.⁶ Although most of the relationship-based financing literature focuses on small firms, new sources of data have allowed researchers to examine the importance of repeated borrowing from the same lender for larger, more transparent firms. Notably, [Dahiya, Saunders, and Srinivasan \(2003\)](#), [Bharath, Dahiya, Saunders, and Srinivasan \(2008, 2007\)](#), and [Ivashina \(2009\)](#) show that

⁶ For a detailed survey of the relationship banking literature, see [Ongena and Smith \(2000\)](#) and [Boot \(2000\)](#).

relationship banking positively affects the lending terms for large syndicated loans.

In the remainder of the article, we formulate our testable hypotheses, describe the data, present our main results, discuss the implications of our findings, and review their robustness. The article concludes with a summary of our conclusions.

1. Testable Hypotheses

The banking literature argues that repeated borrowing from the same lender reduces asymmetric information about firms' quality, thereby improving access to financing and the terms of that financing.⁷ We focus on the information asymmetry between an LBO firm and a lender rather than on the relationship between a borrower and a lender. The former relationship is unique because the formal loan contract is between the portfolio company and the bank, with no direct claim against the financial sponsor (the LBO firm). However, LBO firms are effectively shadow borrowers, as they control the borrower's equity, management, capital structure, and strategic direction. Thus, we argue that banks' relationships with LBO firms are a critical element in lending decisions.

Although LBOs may be large transactions, it would be naïve to think that they are transparent. Most of the targets are held privately by LBO firms, which are typically small organizations with limited public disclosure.⁸ Therefore, banks are likely to acquire valuable private information about LBO firms when underwriting their syndicated bank loans. Examples of private information that may be gathered by banks include information on the LBO firm's due diligence process, its compensation structure, and its internal projections for operating scenarios. A lead bank will communicate with the LBO firm during the bidding process about its plans for the target company, and that bank can evaluate the LBO firm's performance in terms of the execution of its plans. Private information is also likely to be gathered when an LBO firm replaces management, makes acquisitions, or invests in the assets of a previously ignored operational division. The transformational nature of leveraged buyouts suggests that information about leveraged buyout firms and the impact of those firms on borrowers is likely to be "soft" or difficult to transfer (Stein 2002). The bank may gather such soft information on the LBO firm's willingness or ability to contribute additional capital to the company if trouble arises. As long as this information can be reused, each additional loan has a lower marginal cost of monitoring or screening. In a competitive market for bank loans, these lower costs will result in lower interest rates.

⁷ See Boot (2000) for a detailed discussion.

⁸ One of the largest private equity firms, the Blackstone Group, launched an initial public offering in 2007, which was subsequent to the sample period. Notably, disclosure about past investments is quite limited, even in Blackstone's SEC filings. Blackstone never provides information about its track record on a deal-by-deal basis, and it presents most information on an aggregate basis.

Another way in which repeated interactions between banks and LBO firms might reduce the costs of asymmetric information is through reputation building. In the absence of repeated interactions with the lender, equity holders may take actions that endanger the lender's capital, such as investing in excessively risky projects. If an LBO firm expects to seek loans from the bank in the future, it must consider the fact that the bank could punish it in future transactions. This may change the LBO firm's actions *ex ante* (e.g., [Diamond 1991](#); [Chemmanur and Fulghieri 1994](#)). An LBO firm that intends to borrow from a lender in the future may choose safer projects or choose to inject additional capital into a struggling investment. We do not distinguish between these alternative channels. Instead we provide general evidence that lower asymmetric information costs result in lower interest rates.

Our primary measure of the strength of an LBO firm's bank relationship is the percentage of the LBO firm's loans originated by the same lead bank in the previous five years, calculated at time of the transaction. We discuss other proxies for the strength of the lending relationship in more detail in the next section.

Reduced asymmetric information also has implications for loan covenants. Lower monitoring costs or reputation-related concerns should allow for more flexible contracting. Typically, bank loan covenants are set close to default levels (e.g., [Chava and Roberts 2007](#); [Dichev and Skinner 2002](#)). This can reduce potential conflicts of interest by forcing early renegotiation, consistent with Smith and Warner (1979). However, these renegotiations are likely to be costly. With repeated lending relationships, better-informed banks can worry less about potential conflicts of interest and, thus, covenants should be looser.⁹ We refer to our first hypothesis as *reduced information asymmetry* and formalize it as follows:

Hypothesis 1 (H1): *A stronger lending relationship between a lead bank and an LBO firm should result in a lower loan spread and looser financial covenants.*

In addition to underwriting bank loans, banks sell LBO firms many other fee-based services (e.g., M&A advising and securities underwriting). If bank pricing decisions take fees across all products into account, we would expect that the potential for additional business from LBO firms' other investments would improve the terms for any given loan. While theoreticians have little to say about this cross-selling effect on loan pricing, the matter is straightforward to practitioners. As the head of a large leveraged loan desk said: "When pricing the loan, we think about what's coming down the deal pipeline for those guys [the LBO firm]. If we give them good pricing here, we can get paid

⁹ Gârleanu and Zwiebel (2006) predict that under costly renegotiations, contracts will be written to minimize the frequency of future violations, i.e., with looser financial covenants. In general, better information about the borrower should allow for flexibility and discretion in contracting (see [Boot 2000](#) for a discussion of contractual flexibility).

back later.”¹⁰ We see the cross-selling of loans and fee-based services as a complementary channel that affects loan terms. Consistent with Rajan (1992), firms with greater cross-selling potential may have greater bargaining power, which would enable them to extract the surplus from reduced information asymmetry. Thus, we test the role of cross-selling potential *in addition* to the reduced information asymmetry effect.

To the extent that LBO firms prefer looser covenants, cross-selling predicts that banks might offer looser covenants instead of lower spreads. Overall, as the rest of the lending syndicate may not benefit from cross-selling, lead banks may be restricted in their ability to offer lower spreads and looser covenants.¹¹ We refer to our second hypothesis as *cross-selling* and formalize it as:

Hypothesis 2 (H2): *An LBO firm’s potential for paying more bank fees should result in a lower loan spread and may result in looser financial covenants.*

H1 and H2 rely on different mechanisms and are not mutually exclusive. However, evidence for the reduced information asymmetry hypothesis (H1) could actually be evidence for the cross-selling hypothesis (H2). To refute this possibility, we try to distinguish between cross-selling and repeated lending relationships by distinguishing between spread and fees. Repeated lending relationships might reflect higher trust (which facilitates cross-selling) rather than lower costs of information production. In that sense, repeated lending relationships would merely facilitate the commitment to future fees. Thus, to distinguish between H1 and H2, we examine the breakdown of the loan price between spread and fees. Most of the loans in our sample are syndicated. The lead bank keeps only a fraction of the loan, and allocates the rest to syndicate participants or sells it to investors. The upfront fee is typically paid only to the lead bank, and not to the loan syndicate or loan buyers. However, the loan spread is paid to all holders of the loan. While the information benefits of bank relationships accrue to all holders of the loan, profits from cross-selling other financial services are not shared with the rest of the syndicate. If the repeated lending relationship captures cross-selling (H2), then the price effect should be reflected only through the fee that goes directly to the lead bank and not through the loan spread shared by all of the syndicate banks. This leads to our third hypothesis:

Hypothesis 3 (H3): *The higher the information asymmetry benefits of the repeated relationship (H1), the lower the loan spread shared by the lending syndicate.*

¹⁰ In the academic literature, Drucker and Puri (2005) find that loan underwriting increases the probability of winning future advisory business and that future advisory rents are passed back to the borrower through lower interest rates. Yasuda (2005) shows that the lending relationship affects the choice of bond underwriter.

¹¹ Participant banks interact with the borrower less often than the lead bank. However, being part of the lending syndicate allows them to increase their chances of winning other business. Therefore, there could be incentives for all members of a lending syndicate, and not just the lead bank, to accept a reduced interest rate or more relaxed covenant structure.

2. Institutional Background and Data

2.1 Institutional details

Leveraged buyout firms are typically structured as limited partnerships with a fixed life (usually ten years). LBO firms usually disclose as little information as possible about their operations (in our sample period, none of the LBO firms were public). LBO firms bid for companies in an auction organized by an investment bank. As bids are financed mostly with debt, an important component of a bid is a description of the LBO firm's plans for financing the transaction. A final bid will typically include an equity commitment letter from the LBO firm and a debt commitment letter from the lead bank(s). The debt letter will include the terms on which the lead bank commits to arrange the debt financing, including terms for the bank loans that will remain in the capital structure and for any bridge loans to subsequent bond financing. A typical commitment letter will specify a loan price (fees and a spread to LIBOR), loan covenants, and a maximum bond price (subject to a negotiated price or covenant change on unsuccessful syndication). When the target company and its advisors select a winning bidder, they may consider the price (overall transaction value) as well as the way in which the transaction will be financed. The ultimate capital structure of a leveraged buyout typically has numerous levels of seniority, including several debt tranches (60–70% of the capital structure) and equity provided by the LBO sponsor (30–40%).

The amount of senior debt is cyclical, ranging from 25% to 45% of the capital structure from 1997 to 2005. Large LBOs usually include a layer of subordinated debt (10–25%) and may include a layer of mezzanine financing.¹² Debt incurred in conjunction with the LBO typically has recourse only to the borrower. The LBO firm that provides the equity is not liable for any of the debt beyond the amount of equity capital it originally contributed to the LBO. Figure 1 shows the relative amount of different types of debt and equity financing over time for large LBOs.

When the auction winner has been declared, the winning LBO sponsor selects a lead bank. If multiple lead banks are selected, the LBO sponsor allocates the banks' underwriting shares. The lead bank then syndicates the loan, typically keeping 5% to 10% of the tranche. Upfront fees paid to the lead bank for structuring and syndicating the loan average from 1.5% to 2% of the loan amount.

In addition to the fees and interest that banks collect directly from their commitments to finance transactions, banks can hope for future business from the LBO firm and from the company that has been purchased. Non-loan advisory fees are likely to be larger than loan fees (M&A transactions are usually priced at 1% of transaction value and equity offerings at 7% of the equity value).

¹² Capital structure percentage data from Standard & Poor's. Mezzanine financing carries a higher interest rate than subordinated debt and may offer more flexible payment terms, such as the option to pay interest in kind.

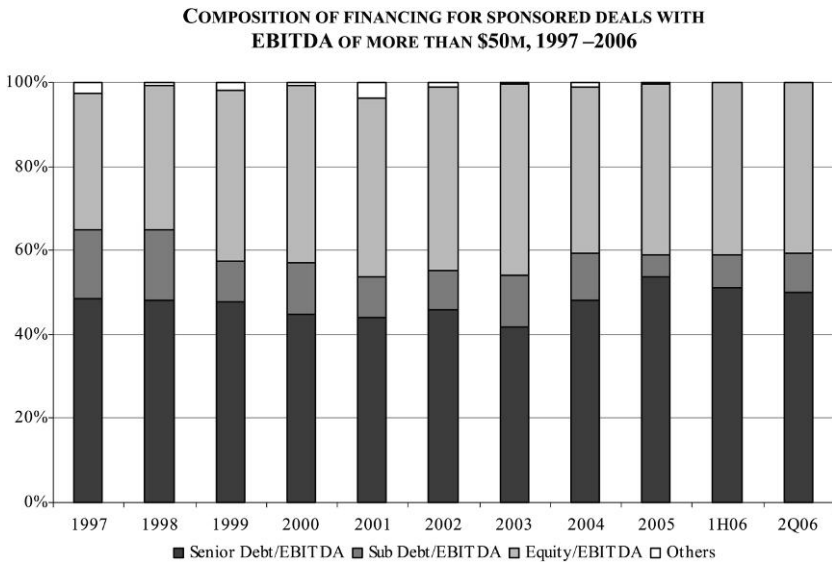


Figure 1
Composition of financing for sponsored deals with EBITDA of more than \$50m, 1997–2006

Many leading LBO firms keep track of the amount of advisory fees paid to each bank and use this information to negotiate loan terms.

2.2 Empirical design and data

The ideal empirical design would be one that allows for a comparison of a sample of the *same* LBO sponsor backing the *same* portfolio firm financed by the *same* bank through the *same* deal structure at the *same* point in time, with banks randomly assigned to transactions and differing only in terms of the intensity of bank relationship. These loans would have identical loan terms so that they could be easily compared. Of course, such an empirical design is impossible. Instead, we control directly for differences in LBO firm, bank, and borrower characteristics, as well as the overall loan market, to the extent that the data is available. We also use an instrument to predict bank relationships. To compare loan terms, we use loan spread and debt to EBITDA, the best represented loan covenant, as dependent variables. We focus on the relevant relationship between an LBO firm and a bank, and not on the relationship between a specific fund within an LBO firm and a bank (e.g., the relevant relationship would be between a bank and KKR, and not between the bank and KKR's 1996 fund).

Data on loan terms come from Reuters' LPC DealScan loan database (DealScan), which provides detailed information as of the loan origination date. DealScan data have been extensively described in previous literature (see, for example, [Ivashina 2009](#)). We include data on the borrower name,

members of the initial lending syndicate, loan size, bank loan covenants, and loan spread.¹³ Loans financing transactions with equity contributions from LBO firms are flagged as sponsored loans, and annotated with the firm and sometimes the fund name. There may be multiple listings in DealScan for each sponsor and company pair as a result of follow-on transactions or refinancings. We select the first chronological listing for an LBO firm and a borrower, so that we include only *one* observation for each sponsor/borrower pair. Of the 1,956 loans that DealScan has registered as sponsor-led transactions between 1993 and 2005, we identify 1,590 that have detailed loan information.¹⁴ A loan may lack data because it was not syndicated or because no information was filed with DealScan. We have no reason to believe that constraining the sample to loans with available data introduces systematic bias in our results. The resulting sample includes 124 LBO firms and 49 lead banks.

Figure 2 shows the volume of sponsored loans in our sample, total institutional sponsored loans, and total LBO transaction values over time. On average, DealScan captured 56% of annual institutional sponsored loan volume from 1996 to 2005.



Figure 2
Sponsored loan volume: US market, 1988–2005

- ¹³ Most of the loans have multiple facilities or tranches within the same loan. The relevant observation for our analysis is a loan; in other words, each loan is counted only once in our sample. Loan size refers to the total size of all of the facilities. All other loan characteristics used in the analysis are set at the loan level and do not change by facility.
- ¹⁴ Approximately 20% of the sample is stand alone revolver lines. Revolver lines allow the borrower flexibility to repay the borrowed amount and provide some financial slack. Revolver lines are costly to the borrower even if the line is unused (undrawn). Thus, they are a function of financing needs and are included in our sample.

Summary statistics for the sample are presented in Table 1, Panel A. The median loan amount is \$136 million, and the mean is \$321 million. This is a large number relative to the median non-sponsored syndicated term loan to a public company in DealScan of \$79 million and the median non-sponsored loan issued to finance an acquisition, stock repurchase, or restructuring of \$106 million. The median sponsored loan maturity is approximately six years, which is one year longer than the median maturity for non-sponsored loans. Selected summary statistics for non-sponsored syndicated loans in DealScan issued to finance an acquisition, stock repurchase, or restructuring are shown in Table 1, Panel B, for comparison. Table 1, Panel C, reports the top ten lender and financial sponsors calculated over the entire sample.

In this article, we focus on the cross-sectional variation of loan terms. We believe that the observed time trend is unlikely to be related to the factors identified in this article. [Ivashina and Sun \(2011\)](#) show that this trend characterizes the leveraged loan market in general and not just leveraged loans associated with LBOs.

Table 1
Summary statistics

Panel A: Summary statistics for sponsored loans, 1993–2005

	Obs.	25%	Median	Mean	75%	StdDev
Loan amount (\$ millions)	1,590	70.0	136.0	321.0	300.0	725.0
Spread (basis points)	1,590	250.0	300.0	314.1	375.0	126.8
Maximum debt to EBITDA	474	4.3	5.5	5.4	6.4	1.7
Number of financial covenants	659	2.0	3.0	2.9	4.0	1.6
Maturity (months)	1,590	60.8	73.0	66.7	85.2	28.4
Number of lenders	1,590	2	4	7.0	8	8.6
Previous loans (\$ millions)	1,590	549.7	1,743.5	3,072.1	4,239.0	3,542.8
Previous loans (same bank) (\$millions)	1,590	50.0	320.0	1,055.5	1,235.0	1,830.8
Bank relationship (%)	1,590	3.8	25.1	31.3	51.0	28.7
Fund size (\$ millions)	1,518	683.0	1,200.0	2,076.1	3,000.0	2,261.3
Previous fee transactions (same bank) (\$millions)	1,590	0.0	0.0	1,380.4	17.0	4,710.2
Previous fee transactions (other banks) (\$millions)	1,590	0.0	320.0	3,428.6	3,338.7	7,111.9
Book leverage	386	0.52	0.72	0.84	0.94	1.63
EBIT to assets	386	0.01	0.03	0.02	0.05	0.23
Sales to assets	386	0.66	1.08	1.24	1.62	0.87
Working capital to assets	386	0.02	0.10	0.08	0.23	0.86
Retained earnings to assets	386	−0.20	−0.02	−0.28	0.05	2.10
No Compustat data (dummy)	1,590	1	1	0.80	1	0.40
New lead bank-borrower (dummy)	1,590	0	1	0.68	1	0.47
Traded loan	1,590	0	0	0.18	0	0.39

Panel B: Characteristics of non-sponsored loans issued to finance an acquisition, stock repurchase or restructuring, 1993–2005

	Obs.	25%	Median	Mean	75%	StdDev
Loan amount (\$ MM)	5,430	45.0	106.2	295.4	300.0	519.6
Spread (basis points)	5,430	125.0	225.0	226.9	300.0	130.5
Maximum debt to EBITDA	1,559	3.0	4.0	4.4	5.3	1.8
Number of financial covenants	2,447	2.0	3.0	3.2	4.0	1.4
Maturity (months)	5,430	36.5	60.9	56.8	73.1	27.3

continued

Table 1
Continued

Panel C: Top-10 lenders and sponsors, 1993–2005

	Lender	Market share		Sponsor	Market Share
1	Bank of America	21.16%	1	Kohlberg Kravis Roberts & Co	6.59%
2	Citibank	8.34%	2	Carlyle Group	6.31%
3	JP Moran Chase	8.27%	3	Blackstone Group	5.64%
4	Bankers Trust	7.55%	4	Apollo Advisors	4.85%
5	Deutsche Bank AG	7.17%	5	Clayton Dubilier & Rice	4.67%
6	Morgan Guaranty Trust	6.23%	6	Bain Capital Partners	4.56%
7	Credit Suisse	6.14%	7	Texas Pacific Group	4.54%
8	Lehman Brothers	6.02%	8	Goldman Sachs	4.17%
9	Goldman Sachs	4.42%	9	Providence Equity Partners	3.88%
10	General Electric Capital Corp	3.85%	10	Thomas H Lee	3.71%

Panel A includes loans originated between 1993 and 2005 sponsored by LBO firms for which the SIC code, maturity, loan amount, sales at close, and spread are available. Panel B includes non-sponsored loans originated between 1993 and 2005 with detailed information available and for which the loan purpose was to finance an acquisition, a stock repurchase, or restructuring. *Spread* is measured in basis points over LIBOR and corresponds to the total cost (interest rate and fees) paid over LIBOR for each dollar drawn down under the loan commitment. *Previous loans* is the dollar value of loans associated with transactions sponsored by the LBO firm in the previous five years. *Bank relationship (amount)* is the dollar value of loans in the previous five years underwritten by the lead bank divided by the total dollar value of all loans sponsored by the private equity firm. *Fund size* is the equity capital of the LBO fund that invested in the borrower. *Previous fee transactions* is the sum of the total enterprise value (TEV) of the deals that each LBO firm participated in (as a buyer or seller) in the past five years that were advised by either the (*same bank*) that underwrote the loan or by all (*other banks*). *Book leverage* is book debt divided by total assets. *EBIT to assets* is earnings before interest and taxes divided by total assets. *Sales to assets* is net sales divided by total assets. *Working capital to assets* is working capital divided by total assets. *Retained earnings* is working capital divided by total assets. *No Compustat data* is a dummy variable equal to one if no Compustat data are available for the borrower. *New lead bank-borrower* is a dummy variable equal to one if the lead bank is not the lead bank on the borrower's (i.e., the portfolio company's) previous syndicated loan or if there was no previous syndicated loan. *Maximum debt to EBITDA*, *Number of financial covenants*, and *Fund size* are computed where data are available. *Share* is the dollar-weighted share of the sponsor or lender in the entire sample.

2.3 Dependent variables

Our primary dependent variables are *Spread* and *Maximum Debt to EBITDA*. *Spread* measures the interest rate that the borrower pays on the loan. It is calculated in basis points and corresponds to the total cost (interest rate and fees) paid over LIBOR for each dollar drawn down under the loan commitment. The median spread to LIBOR in the sample is 300 basis points, with a 25th-percentile loan at 250 basis points and a 75th-percentile loan at 375 basis points.¹⁵

As discussed in the previous section, the loan covenant structure is an essential element of a loan contract. It is harder to assign a monetary value to more lenient covenants, but such covenants provide financial flexibility, which may prevent bankruptcy in adverse economic conditions. Although we would like to have an “apples-to-apples” comparison of the tightness of covenants across loans, covenant information disclosed in DealScan is relatively limited.

¹⁵ As a point of comparison, the median statistics for similar, non-sponsored loans are loan size of \$106 million, a spread of 225 basis points, and maximum debt to EBITDA of 4.0x (see Table 1, Panel B).

In order to evaluate the tightness of loan covenants, we focus our analysis on the *Maximum debt to EBITDA* covenant. This covenant offers several advantages. First, industry participants (leveraged buyout firms and syndicated lenders) identify this as the most important covenant.¹⁶ Second, consistent with this anecdotal evidence, *Maximum debt to EBITDA*, used in 29% of the loans, is the covenant best represented in our sample. Finally, while covenants may be renegotiated over the course of the loan's term, total debt to cash flow is likely to be critical for a highly leveraged company because it limits the amount of debt used to finance the initial transaction (Roberts and Sufi 2008). Evidence of the importance of this covenant to LBO transaction financing is found in the fact that, of the DealScan observations with covenant information, maximum debt to EBITDA is included in 82% of the LBO financings, while it is included only for 45% of the loans in the overall DealScan database.

We would prefer to measure the degree to which the financial covenants are binding rather than their absolute level. However, as most of these companies are private after the leveraged buyout, no detailed financial information is available. In addition, the actual maximum debt to EBITDA ratio is difficult to calculate because it could be defined in many different ways. Previous studies, including Dichev and Skinner (2002) and Drucker and Puri (2007), overcome this problem by looking at the minimum net worth and current ratio covenants. These financial ratios are less ambiguous, but they are rarely used in LBO financings.¹⁷

Instead of directly calculating covenant slack as the difference between the covenant and the actual financial ratio, we control for the company's financial results by using book leverage, EBIT to assets, and sales to assets ratios as independent variables to the extent that we have the requisite information. We include a dummy variable equal to 1 (*No Compustat*) for observations without Compustat data.

Cross-sectional heterogeneity in defining maximum debt to EBITDA is likely to introduce measurement error because PE firms' and banks' definitions of EBITDA vary systematically. However, we expect accounting definitions to be stable within PE firms, banks, and/or industries. Thus, we verify that our results are robust to the inclusion of LBO firm, bank, and industry fixed effects.

The median maximum debt to EBITDA covenant in the sample is 5.5x (mean of 5.4x). This is much higher than the median maximum debt to EBITDA covenant of 4.0x in non-sponsored loans issued to finance acquisitions, stock repurchases, or restructurings. Assuming that LBO firms substitute debt for

¹⁶ See, for example, "What Private Equity Firms Want in a Lender," *Commercial Lending Review*, 01/2007.

¹⁷ We also looked at the number of covenants as an alternative dependent variable: The bank relationship coefficient was negative (ranging from -0.4 to -0.5) and significant at the 5% to 10% level (results available upon request). Theoretically, however, the association between the bank relationship and the number of covenants is unclear. It is possible that a better understanding of the borrower and the LBO firm could be associated with a larger number of covenants or more detailed covenants.

equity up to the covenant limit, moving from the median debt to EBITDA of 5.5x to the 75th-percentile debt to EBITDA of 6.4x increases the average equity return from 25% to 35%.¹⁸ Note that the so-called “covenant-lite” loans did not become popular until 2006 and are, therefore, not included in our sample. For conservatism, the *Maximum debt to EBITDA* analysis excludes deals with omitted maximum debt to EBITDA covenant information, although it is robust to assuming that the omitted covenants are not binding.

2.4 Explanatory variables

In order to test our first hypothesis (H1), we need to measure the bank relationship—the fixed cost that a lead bank has invested in collecting information about the LBO firm that can be reused in future transactions or the investment of an LBO firm in its reputation with a lead bank. To test H1, we seek to define a measure of repeated interactions that would capture the duration and intensity of the relationship.¹⁹

We follow Bharath, Dahiya, Saunders, and Srinivasan (2007), and use a scale-independent measure of the LBO firm’s relationship with its lead bank, *Bank relationship*. For a given loan, *Bank relationship* is the dollar value of loans in the previous five years underwritten by the same lead bank divided by the total dollar value of all loans sponsored by the LBO firm. When there are multiple lead banks, the highest *Bank relationship* value is selected. For example, if Bank of America (bank relationship of 0.20) and Citibank (bank relationship of 0.35) are co-leads, the bank relationship measure used in the estimation will be 0.35. The average *Bank relationship* in the sample is 31%, and the standard deviation is 29%. This variation is present both within and across LBO firms. The average standard deviation within LBO firms is 28%. Measuring bank relationships for traditional borrowers presents several challenges due to the fact that variation in loan maturity and, thus, the frequency of repeated interactions between the borrower and the bank capture credit characteristics of the borrower. As we define relationship variables on the sponsor level, and as sponsors repeatedly seek loan financing as they acquire new companies for reasons unrelated to the current loan, this problem is not present in our measures.

Our primary measure of the bank relationship is based on the past loan amounts underwritten by the same bank. For illustrative purposes, we calculate *Bank relationship (N)* based on the number of loans sponsored by the PE firm and underwritten by the same bank in the previous five years divided by the total number of loans sponsored by the PE firm. We believe that the measure based on loan amount captures reduced information asymmetry better than

¹⁸ This assumes total enterprise value growth of 6%, debt paydown of 25%, exit in four years, and an initial equity contribution of 30%.

¹⁹ Reputation and monitoring both result in reduced information asymmetry. We do not distinguish between these two channels in this article.

Bank relationship (N) because, unlike the number of loans, the loan amount captures banks' dollar exposure to an LBO firm. Other measures of the bank relationship that are not scale independent, such as the number or amount of previous loans, could conflate the measure of the bank relationship with a measure of firm quality (better LBO firms stay in business longer and purchase increasingly large companies).

Consistent with Bharath, Dahiya, Saunders, and Srinivasan (2007), we restrict bank-related measures to transactions within the past five years. Five years is a reasonable period of time during which a monitoring investment could yield information relevant to an LBO firm's current operations or its reputation. We estimate similar results with measures calculated from all transactions since the LBO firm's inception. We analyze only loans arranged after 1993 so that we have five years of data, and we drop LBO firms' first investments, which all would have a *Bank relationship* value of zero. For the initial transactions of the four LBO firms that were established by a founding partner leaving an original firm, we assigned the *Bank relationship* level of their original firm.

In order to control for the LBO firms' characteristics, we augment the DealScan data with information from the Private Equity Performance Monitor, which offers details on over 1,000 PE firms, including fund-by-fund performance and fund size information. The database is compiled from limited partners, general partners, and public sources, and is described in more detail in Lerner, Schoar, and Wong (2006). In addition, we use information provided by S&P's Capital IQ database on fund size to further supplement this database. We measure *Fund size* as the logarithm of the amount of the equity capital of the fund associated with that investment. As LBO firm size is correlated with firm quality, we use fund size as a control for LBO firm quality.²⁰ As would be expected given the large amount of the bank loans, the sample includes mostly larger LBO funds (median fund size is \$1.2 billion).²¹

Fund size is available for only 1,518 observations. We therefore also calculate *Previous loans* as the log of one plus the dollar value of loans associated with transactions sponsored by the LBO firm in the previous five years. *Fund size* and *Previous loans* have a correlation of 0.70, and *Previous loans* is available for all observations. The median value of *Previous loans* is \$1.7 billion, and the mean is \$3.1 billion.

To test our second hypothesis (H2), we need to measure cross-selling opportunities from non-loan businesses. Ideally, we would measure the lead bank's

²⁰ As Metrick and Yasuda (2010) observe, unlike the venture capital industry, leveraged buyouts are scalable. LBO firms that experience success tend to raise increasingly larger funds to invest in larger targets.

²¹ In order to do this mapping, we make some assumptions. For example, because KKR's 1996 fund was raised in 1996 and KKR's Millennium Fund was raised in 2000, we attribute all investments made between 1996 and 2000 to KKR's 1996 fund. As fund sizes grew during this time period, this metric underestimates firm size when investments are made in a fund's later years. The last investment in a firm's \$5 billion fund and the first investment in its \$10 billion fund may be a day apart. It is unlikely that there are systematic differences in the timing of firms' investments that would cause bias, but this measurement error may add noise.

expectation of future business from an LBO firm. However, as we cannot measure such expectations, we assume that past business predicts future business.²² *Previous fee transactions (same bank)* is the logarithm of one plus the sum of the total enterprise value of the transactions associated with the LBO firm advised by the lead bank in the previous five years. The correlation with the total enterprise value of the transactions associated with the LBO firm advised by the lead bank in the *next* five years is 0.66, which increases our confidence that this is a good measure for predicting expectations of future business.²³

Transaction data comes from a hand-match of LBO firm names and lead bank names to S&P's Capital IQ database. This procedure allows us to compile information on all of the transactions in which the LBO firm has participated as a seller or a buyer. When there are multiple lead banks, the bank with the highest cross-selling value is selected. The mean level of *Previous fee transactions (same bank)* is \$1.4 billion, and the median is 0. We run similar unreported specifications using the number of previous fee transactions advised by the same bank because not all Capital IQ transactions have information on enterprise value and because the amount measure may be biased toward LBO sponsors that acquire public companies. We arrive at similar results.

When testing H2, it is difficult to find a measure of future fees that is not directly correlated with LBO firm size, as fees, transaction size, and LBO firm size are all directly correlated. Unlike with reduced information asymmetry (H1), where participating in a greater percentage of an LBO firm's investments means that a bank may learn disproportionately more about that LBO firm, each transaction dollar translates directly into fee income. Therefore, it is doubly important to control for LBO firm size here. We do so using *Previous fee transactions (other banks)*, the logarithm of one plus the sum of the total enterprise value of the transactions associated with the LBO firm advised by other banks in the previous five years. We also control for firm size with *Previous loans* and *Fund size*.

As borrower quality affects loan terms, we attempt to control for differences in credit risk using borrower characteristics. Following previous literature (Ivashina 2009; Carey and Nini 2007), we control for (i) *Deal amount*—the logarithm of the total loan amount; (ii) *Maturity*—maturity (in months); (iii) *Unsecured*—a dummy variable equal to one if the loan is not secured; and (iv) *Number of lenders*—the number of lenders in the syndicate. In addition, we include credit ratings' fixed effects for rated companies—a dummy variable

²² In an unreported regression, we reexamine the result measuring bank's expected income from cross-selling based on future transactions. The choice of the variable does not change the results.

²³ In an unreported analysis, we estimate a logit model of underwriter selection following Drucker and Puri (2005), where the dependent variable is a dummy equal to one if the bank is selected as the advisor for any merger, acquisition, or public offering involving the leveraged buyout firm and its investments. Our findings confirm that previous fee transactions are predictive of future underwriting selection. While we do not know the exact fee amount, fees are usually a fixed percentage of transaction value and thus transaction value serves as a good proxy for fees. The results are available from the authors upon request.

equal to one for each category of the S&P loan rating (i.e., with a separate dummy for B, B+, and B- ratings). Most of the rated loans are BB through B+. In addition, for the subset of the borrowers that were public companies immediately prior to the leveraged buyout and for which Compustat data are available (24% of the overall sample), we control for *Book leverage*, measured as book debt to assets, and *ROA*, measured as EBIT to assets. We include a dummy variable equal to one for borrowers for which Compustat data are not available (*No Compustat*). Thus, the coefficients on the Compustat-based performance measures correspond to the subsample of the firms for which data are available. In addition, we control for heterogeneity in banks with the ordinal rank of the bank calculated using market share based on the five-year rolling window (*Lender rank*).

While practitioners describe the choice of bank as solely that of the LBO firm, LBO firms are not the only entities with bank relationships. Borrowers (i.e., portfolio companies) may have a pre-existing relationship with the lead bank. We want to distinguish between the borrowers' relationships and the LBO firms' relationships because we focus on the importance of only the LBO firms' bank relationships. Following [Bharath, Dahiya, Saunders, and Srinivasan \(2007\)](#), we use a dummy variable to control for *New lead bank-borrower*, which is equal to one when the lead bank on the sponsored loan was not the lead bank on the borrower's previous syndicated loan or when the borrower did not have a previous syndicated loan.²⁴ On average, 68% of the LBO sponsored loans do not use the borrower's previous relationship bank.

We also include *Traded loan*, a dummy variable equal to 1 if the loan is ultimately sold in the secondary loan market and 0 otherwise. We use this variable as a proxy for higher expected liquidity, which, according to [Drucker and Puri \(2009\)](#), is likely to reduce loan spread and result in higher leverage ratios.²⁵

Previous literature has shown that LBO-sponsor financing varies with market conditions ([Kaplan and Stein 1993](#)). In practice, a limited number of LBOs are financed when funding conditions are tight, so there is a limited sample of transactions in tight credit markets. To control for time variations in market conditions, we control for year and industry fixed effects throughout the analysis.

2.5 Instrumental variables

It is possible that bank selection is correlated with the unobservable quality of the deal. Repeated lending relationships might be associated with more

²⁴ We estimate similar results using continuous measures of the borrowers' bank relationship and previous loans as controls (calculated at the borrower/bank level rather than the leveraged buyout firm/bank level).

²⁵ [Drucker and Puri \(2009\)](#) also find that loans that are likely to be traded on the secondary market are associated with higher lending relationships between the bank and the borrower. In our sample, the correlation between *New lead* and *Traded loan* is -0.08 . This confirms that the effect we measure is unlikely to be related to the liquidity effect estimated in Drucker and Puri's paper.

favorable loan terms if LBO firms take their unobservably good loans to the same bank but their unobservably bad loans to different banks. Alternatively, LBO firms may take their unobservably bad loans to their relationship bank, which would negatively bias our estimates. Therefore, we use an instrument that predicts bank relationship but is orthogonal to loan terms.

Despite the benefits of relationship banking, it is not unusual for borrowers to use “non-relationship” banks. [Ongena and Smith \(2001\)](#), [Houston and James \(2001\)](#), [Farinha and Santos \(2002\)](#) and, more recently, [Gopalan, Udell, and Yerramilli \(2008\)](#) investigate this phenomenon in a non-LBO context. Their results suggest that firms change banks in order to improve their access to better credit terms. Interviews with practitioners in the LBO industry suggest that financial sponsors maintain competition in loan pricing by building relationships with several banks. We use this rationale for bank-switching behavior to instrument for lending relationships.²⁶

Ideally, there would be an exogenous rule for the bank-selection process. For example, every time a PE firm’s relationship with a single bank is higher than some fixed threshold, the PE firm selects a new bank to maintain its bargaining power. Such a simplistic rule is unlikely to exist in practice, and we do not observe any clear discontinuities in bank-switching behavior in the data. However, there is a strong negative relationship between the lagged relationship-concentration measure and the likelihood of selecting a new (non-relationship) bank. Our instrument captures this dynamic. For each deal in the sample, we construct a measure of the PE firm’s highest and second-highest bank relationships *prior* to the specific transaction. The measures are computed in the same way as is our main relationship measure and are unrelated to the selection of the bank for the analyzed deal. The strategy is to identify cases in which the lender was likely chosen to reduce overexposure to a relationship bank. We calculate the difference between the PE firm’s highest and second-highest bank relationships (*Difference in lagged bank relationship*). For example, if a private equity firm’s highest relationship bank has a 60% share and its next-highest relationship bank has a 20% share, the difference would be 40%. The larger the disparity in deal allocation between the top two banks, the more likely that the PE firm will select a new lender. Indeed, when this difference is high, PE firms are likely to select banks with lower relationships. This measure is unlikely to be correlated with unobservable LBO characteristics and is based only on lagged bank relationship measures, so that it is not dependent on any features of the bank actually selected for the transaction. In addition, we include the PE firm’s second-highest bank relationship (*Lagged bank relationship 2nd*) to control for the level effect.

The results for the first stage of the instrumental variable specification are presented in the appendix. The greater the difference between the first and second relationship banks, the more the PE firm may need to maintain its

²⁶ We thank an anonymous referee for this suggestion.

bargaining power by diversifying, leading to the selection of a lower-relationship bank and, thus, a negative coefficient in the first stage. The higher the PE firm's relationship with its second-highest relationship bank, the less of a problem the selection of a high-relationship bank becomes, which leads to a positive coefficient in the first stage.

3. Cross-sectional Analysis of Sponsored Loans

3.1 Results

Table 2 shows the results of a regression of loan spreads on measures of bank relationship, which includes controls for loan characteristics. This tests hypothesis H1, which states that stronger bank relationships should result in lower spreads. As predicted, a higher *Bank relationship* between a lead bank and an LBO firm is associated with lower loan spreads (specification 3). This result is robust to the inclusion of controls for firm size as measured by *Previous loans* and *Fund size* (specifications 4 and 5). A one-standard-deviation increase in the bank relationship strength engenders a 7.5-basis-point decrease in spread (specification 4). At the extreme, if an LBO firm moves from a lead bank that financed none of its previous LBOs to a lead bank that financed all of its previous LBOs, we estimate a 26-basis-point lower spread. As predicted, controls for LBO firm size are negative, although they are significant only at the 10% level.

Specification 7 presents the results of the second stage of the two-stage least squares (2SLS) specification using the instrumented bank relationship as an explanatory variable. The size of the coefficient more than doubles, and the results remain significant at the 5% level. The increase in the economic significance of the coefficient on the relationship-bank variable is consistent with statements from practitioners that they bring riskier, more complex deals to their relationship banks. Thus, as we are unable to fully control for deal complexity, the ordinary least squares (OLS) coefficients are likely to be biased toward zero. We also present results for *Bank relationship (N)* (the bank relationship calculated from the number of loans; specification 6). The coefficient on *Bank relationship (N)* is, as expected, negative, and it is significant at the 10% level.

The first specification of Table 2 includes only controls for the observable characteristics of the loan and borrower. The signs on the coefficients are consistent with previous studies.²⁷ We refer to these controls as the benchmark

²⁷ The negative relationship between *No loan rating*, with a dummy equal to one if the loan is not rated, *Unsecured*, with a dummy equal to one if the loan is not collateralized, and *Spread* is likely to be related to an unobservable selection process. For example, if only very good companies do not need to be rated to receive loan financing, we would observe a negative coefficient. Similarly, if only bad companies are required to offer collateral to secure loans, we would observe a negative relationship.

Table 2
Loan spread

	(1) OLS Coeff.		(2) OLS Coeff.		(3) OLS Coeff.		(4) OLS Coeff.		(5) OLS Coeff.		(6) OLS Coeff.		(7) 2SLS Coeff.		(8) OLS Coeff.	
Bank relationship (amount)	—		−35.07 (3.34)	***	−25.47 (2.58)	***	−26.15 (2.63)	***	−23.62 (2.31)	**	—		−53.35 (1.96)	**	−29.19 (1.65)	*
Bank relationship (N)	—		—		—		—		—		−20.87 (1.97)	**	—		—	
Previous loans	—		—		—		−4.27 (2.16)	**	—		−4.41 (2.20)	**	−4.44 (2.28)	**	0.33 (0.08)	
Fund size	—		—		—		—		−12.68 (3.80)	***	—		—		—	
Unsecured	−47.72 (3.39)	***	—		−48.77 (3.48)	***	−47.24 (3.40)	***	−40.66 (2.81)	***	−46.93 (3.36)	***	−48.31 (3.58)	***	−30.70 (1.38)	
Number of lenders	−0.65 (1.95)	*	—		−0.64 (1.91)	*	−0.63 (1.88)	*	−0.61 (1.77)	*	−0.62 (1.86)	*	−0.62 (1.91)	*	−0.66 (1.49)	
Deal amount	−19.68 (5.10)	***	—		−18.87 (4.95)	***	−17.27 (4.48)	***	−14.28 (3.62)	***	−17.66 (4.55)	***	−16.35 (4.19)	***	−13.48 (2.08)	**
Maturity	1.16 (10.66)	***	—		1.14 (10.45)	***	1.14 (10.40)	***	1.11 (9.84)	***	1.15 (10.42)	***	1.12 (10.37)	***	1.27 (5.19)	***
Book leverage	−29.45 (3.07)	***	−31.60 (3.17)	***	−29.10 (3.04)	***	−28.18 (2.93)	***	−35.25 (3.46)	***	−28.77 (3.00)	***	−27.77 (3.00)	***	−16.77 (1.47)	
EBIT to assets	−16.79 (0.28)		19.36 (0.30)		−18.29 (0.31)		−19.22 (0.32)		−37.92 (0.62)		−17.64 (0.29)		−20.86 (0.37)		−32.31 (0.56)	
Sales to assets	12.71 (2.12)	**	12.53 (2.03)	**	12.39 (2.06)	**	12.44 (2.08)	**	12.46 (2.06)	**	12.72 (2.12)	**	12.10 (2.08)	**	14.62 (2.00)	**
Working capital to assets	−41.60 (2.36)	**	−41.19 (2.23)	**	−39.46 (2.24)	**	−38.76 (2.18)	**	−21.88 (1.19)	**	−39.95 (2.25)	**	−36.44 (2.10)	**	−29.99 (1.55)	
Retained earnings to assets	−14.79 (1.86)	*	−21.09 (2.30)	**	−15.46 (2.00)	**	−15.04 (1.90)	*	−27.39 (2.65)	***	−15.06 (1.89)	*	−15.75 (2.07)	**	−10.70 (1.51)	
No Compustat data	25.46	**	32.89	***	25.06	**	25.03	***	25.05	***	25.33	***	24.60	***		

continued

Table 2
continued

	(1) OLS Coeff.		(2) OLS Coeff.		(3) OLS Coeff.		(4) OLS Coeff.		(5) OLS Coeff.		(6) OLS Coeff.		(7) 2SLS Coeff.		(8) OLS Coeff.
New lead bank - borrower	(2.57) 18.29 (2.83)	***	(3.30) 24.32 (3.66)	***	(2.57) 18.14 (2.81)	***	(2.59) 18.03 (2.79)	***	(2.59) 17.94 (2.68)	***	(2.60) 18.06 (2.80)	***	(2.66) 17.87 (2.85)	***	– 3.11 (0.28)
No loan rating	–27.36 (1.84)	*	–41.52 (2.71)	***	–25.63 (1.71)	*	–23.64 (1.58)		–31.71 (2.05)	**	–25.54 (1.72)	*	–21.72 (1.48)		–64.35 (1.49)
Traded loan	12.10 (1.48)		16.95 (2.21)	**	12.71 (1.55)		13.34 (1.62)		10.73 (1.29)		13.15 (1.60)		14.01 (1.76)	*	26.33 (1.76)
Lender rank	0.28 (1.91)	*	0.30 (2.05)	**	0.20 (1.40)		0.17 (1.14)		0.14 (0.95)		0.20 (1.37)		0.09 (0.54)		0.23 (0.66)
Fixed effects:															
Year	yes		yes		yes		yes		yes		yes		yes		yes
Industry (2 digit SIC code)	yes		yes		yes		yes		yes		yes		yes		yes
Credit rating	yes		yes		yes		yes		yes		yes		yes		yes
Observations	1,590		1,590		1,590		1,590		1,518		1,590		1,590		386
Adj. R-squared	0.35		0.30		0.36		0.36		0.37		0.36		0.36		0.53

The dependent variable is *Spread* measured in basis points. It corresponds to the total cost (interest rate and fees) paid over LIBOR for each dollar drawn down under the loan commitment. The sample includes observations from 1993 through 2005. *Bank relationship (amount)* is the dollar value of loans in the previous five years underwritten by the lead bank divided by the total dollar value of all loans sponsored by the PE firm. *Bank relationship (N)* is the number of loans in the previous five years underwritten by the lead bank divided by the total number of loans sponsored by the PE firm. *Fund size* is the logarithm of the amount of the equity capital of the LBO fund that invested in the borrower. *Previous loans* is the log of one plus the dollar value of loans associated with transactions sponsored by the LBO firm in the previous five years. *Deal amount* is the logarithm of the total loan amount. *Maturity* is the maturity (in months). *Unsecured* is a dummy variable equal to 1 if the loan is not secured. *Number of lenders* is the number of lenders in the syndicate. *Lender rank* is the ordinal rank of the bank calculated using market share based on the five-year rolling window. *Traded loan* is a dummy variable equal to 1 if the loan is ultimately sold in the secondary loan market and 0 otherwise. *New lead bank-borrower* is a dummy variable equal to one if the lead bank was not the lead bank on the borrower's previous syndicated loan or if there was no previous syndicated loan. *Book leverage* is book debt divided by total assets. *EBIT to assets* is earnings before interest, taxes, and interest divided by total assets. *Sales to assets* is net sales divided by total assets. *Working capital to assets* is working capital divided by total assets. *Retained earnings* is working capital divided by total assets. *No Compustat data* is a dummy equal to 1 if the borrower is not in Compustat (1,204 observations). Regression (8) includes only companies for which Compustat data are available. *No loan rating* is a dummy equal to 1 if the loan is not rated. Each regression includes year, industry, and credit rating fixed effects. See Table 1 for data availability. Regressions (1)–(6) and (8) are OLS specifications. Regression (7) is the second stage of the 2SLS specification where *Bank relationship (amount)* is the instrumented variable. See the appendix for the first stage of the 2SLS specification. Standard errors are clustered at the bank and LBO firm level. *t*-statistics are reported in parentheses. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

controls, and they are included in the subsequent specifications. In addition, all specifications include fixed effects for the industry (two-digit SIC codes), the loan origination year, and the loan credit rating (if available).²⁸ Throughout the article, standard errors are clustered on the (bank x PE firm) level. As many of the control variables are endogenous, we estimated the same specifications without deal size, collateral, maturity, and number of lenders, with similar results.

We confirm the predictions of our first hypothesis with similar specifications by regressing *Maximum debt to EBITDA* on loan characteristics and bank relationship (Table 3).²⁹ Even though the sample size drops dramatically due to the limited availability of covenant data, we arrive at similar results for the key variable of interest, with opposite signs for the spread estimation. A one-standard-deviation increase in *Bank relationship* results in a 0.21 point higher maximum debt to EBITDA covenant (Table 3, specification 4) when we control for *Previous loans*. At the extreme, if an LBO firm moves from a lead bank that financed none of its previous LBOs to a lead bank that financed all of its previous LBOs, we would expect a 0.73 point higher maximum debt to EBITDA covenant. The results again support H1, suggesting that informational asymmetry is reduced for firms that have a closer relationship with the lead bank. We find similar results when controlling for firm size with *Fund size* (specification 5). The coefficient on *Bank relationship (N)* after controlling for firm size is significant at the 10% level. Again, the magnitude of the coefficient estimated with the instrumented bank relationship variable is more than twice that in the previous specifications and the coefficient is significant at the 5% level (specification 7).

We next test our second hypothesis (H2), which focuses on cross-selling. Banks appear to reward LBO firms for cross-selling opportunities with lower spreads. Borrowers have lower spreads and looser covenants when the lead bank advised the LBO firm in previous transactions (Table 4, Panels A and B). A one-standard-deviation increase in previous advisory business is associated with a 12-basis-point decrease in the spread. In addition, cross-selling opportunities appear to result in looser covenants. A one-standard-deviation increase in *Previous fee transactions (same bank)* leads to a 0.20-point increase in the maximum debt to EBITDA covenant (specification 2), although the significance falls to the 5% level. We continue to find a significant relationship for past advisory business after controlling for LBO firm size with *Previous loans* (Table 4, Panels A and B, specification 2) and *Fund size* (Table 4, Panels A and B, specification 3). The results are even more dramatic in the 2SLS specifications.

²⁸ We also run unreported specifications with controls for loan origination quarter, with similar results.

²⁹ We follow a conservative approach and constrain the regression analysis to the sample where maximum debt to EBITDA is reported. Therefore, we assume that maximum debt to EBITDA is missing if it is not included in the DealScan observation. Our results are robust to assuming that maximum debt to EBITDA is not binding when it is not reported.

Table 3
Maximum debt to EBITDA

	(1) <i>OLS</i> Coeff.	(2) <i>OLS</i> Coeff.	(3) <i>OLS</i> Coeff.	(4) <i>OLS</i> Coeff.	(5) <i>OLS</i> Coeff.	(6) <i>OLS</i> Coeff.	(7) 2SLS Coeff.	(8) <i>OLS</i> Coeff.						
Bank relationship (amount)	—	0.77 (2.93)	***	0.75 (2.89)	***	0.73 (2.81)	***	0.69 (2.48)	**	—	2.09 (2.22)	**	0.95 (1.85)	*
Bank relationship (N)	—	—		—		—		0.56 (1.82)	*	—	—	—	—	
Previous loans	—	—		—	***	0.18 (3.88)	—	0.19 (4.10)	***	0.17 (3.57)	***	0.25 (2.62)	***	
Fund size	—	—		—		0.17 (3.80)	**	—	—	—	—	—	—	
Unsecured	−0.43 (1.11)	—		−0.48 (1.22)		−0.52 (1.35)		−0.50 (1.15)		−0.52 (1.39)		−0.62 (1.58)	0.13 (0.24)	
Number of lenders	−0.01 (1.08)	—		−0.01 (1.02)		−0.01 (0.89)		−0.01 (0.97)		−0.01 (0.94)		0.00 (0.82)	−0.02 (1.37)	
Deal amount	0.22 (2.38)	**	—	0.19 (2.03)	**	0.15 (1.54)		0.15 (1.38)		0.16 (1.60)		0.10 (0.98)	0.25 (1.39)	
Maturity	0.03 (4.68)	***	—	0.03 (4.78)	***	0.03 (4.68)	***	0.03 (4.69)	***	0.03 (4.59)	***	0.03 (5.16)	0.03 (2.62)	***
Book leverage	−0.08 (0.31)	−0.25 (0.88)		−0.14 (0.53)		−0.14 (0.53)		−0.21 (0.80)		−0.11 (0.41)		−0.25 (0.96)	−0.06 (0.11)	
EBIT to assets	−0.91 (0.34)	−0.67 (0.24)		−0.82 (0.31)		−0.98 (0.39)		−0.75 (0.29)		−1.07 (0.43)		−0.81 (0.35)	−0.71 (0.29)	
Sales to assets	−0.29 (2.53)	**	−0.27 (2.27)	**	−0.27 (2.27)	−0.24 (2.07)	**	−0.26 (2.20)	**	−0.25 (2.18)	**	−0.19 (1.69)	−0.28 (1.56)	
Working capital to assets	−0.67 (0.81)	−0.89 (1.16)		−0.80 (1.00)		−0.91 (1.16)		−0.71 (0.89)		−0.86 (1.07)		−1.13 (1.71)	−1.46 (1.05)	
Retained earnings to assets	−0.97 (2.67)	***	−1.19 (2.86)	***	−1.01 (2.82)	−0.95 (2.74)	***	−1.24 (4.28)	***	−0.91 (2.61)	***	−1.04 (3.23)	−0.76 (1.64)	
No Compustat data	−0.19	−0.11		−0.20		−0.21		−0.24		−0.21		−0.23		

continued

Table 3
Continued

	(1) OLS Coeff.		(2) OLS Coeff.		(3) OLS Coeff.		(4) OLS Coeff.		(5) OLS Coeff.		(6) OLS Coeff.		(7) 2SLS Coeff.		(8) OLS Coeff.
New lead bank-borrower	(0.56) −0.46 ***		(0.32) −0.61 ***		(0.59) −0.46 ***		(0.65) −0.43 ***		(0.72) −0.45 ***		(0.64) −0.43 ***		(0.79) −0.44 ***		− 0.37 (1.50)
No loan rating	0.70 (3.27) (1.55)		0.70 (4.23) (1.70)	*	0.64 (3.32) (1.41)		0.53 (3.16) (1.19)		0.54 (3.11) (1.21)		0.59 (3.14) (1.30)		0.42 (3.47) (1.03)		−5.06 (2.39)
Traded loan	0.08 (0.48)		0.54 (3.23)	***	0.03 (0.17)		0.01 (0.07)		0.08 (0.47)		0.03 (0.19)		−0.08 (0.48)		−0.18 (0.58)
Lender rank	0.00 (0.75)		− (0.75)		0.00 (1.12)		0.00 (1.19)		0.01 (1.29)		0.00 (0.96)		0.01 (1.95)	*	0.01 (0.81)
Fixed effects:															
Year	yes		yes		yes		yes		yes		yes		yes		yes
Industry (2 digit SIC code)	yes		yes		yes		yes		yes		yes		yes		yes
Credit rating	yes		yes		yes		yes		yes		yes		yes		yes
Observations	474		474		474		474		448		474		474		227
Adj. <i>R</i> -squared	0.51		0.46		0.52		0.54		0.55		0.53		0.50		0.62

The dependent variable is the *Maximum total debt to EBITDA* covenant set at loan origination. The sample includes observations from 1993 through 2005. *Bank relationship (amount)* is the dollar value of loans in the previous five years underwritten by the lead bank divided by the total dollar value of all loans sponsored by the PE firm. *Bank relationship (N)* is the number of loans in the previous five years underwritten by the lead bank divided by the total number of loans sponsored by the PE firm. *Fund size* is the logarithm of the amount of the equity capital of the LBO fund that invested in the borrower. *Previous loans* is the log of one plus the dollar value of loans associated with transactions sponsored by the LBO firm in the previous five years. *Deal amount* is the logarithm of the total loan amount. *Maturity* is the maturity (in months). *Unsecured* is a dummy variable equal to 1 if the loan is not secured. *Number of lenders* is the number of lenders in the syndicate. *Lender rank* is the ordinal rank of the bank calculated using market share based on the five-year rolling window. *Traded loan* is a dummy variable equal to 1 if the loan is ultimately sold in the secondary loan market and 0 otherwise. *New lead bank-borrower* is a dummy variable equal to one if the lead bank was not the lead bank on the borrower's previous syndicated loan or if there was no previous syndicated loan. *Book leverage* is book debt divided by total assets. *EBIT to assets* is earnings before interest, taxes, and interest divided by total assets. *Sales to assets* is net sales divided by total assets. *Working capital to assets* is working capital divided by total assets. *Retained earnings* is working capital divided by total assets. *No Compustat data* is a dummy equal to 1 if the borrower is not in Compustat (247 observations). Regression (8) includes only companies with Compustat data available. *No loan rating* is a dummy equal to 1 if the loan is not rated. See Table 1 for data availability. Each regression includes year, industry, and credit rating fixed effects. Regressions (1)-(6) and (8) are OLS specifications. Regression (7) is the first stage of a 2SLS specification where the dependent variable is *Bank relationship (amount)*. Regression (7) is the second stage of the 2SLS specification where *Bank relationship (amount)* is the instrumented variable. See the appendix for the first stage of the 2SLS specification. Standard errors are clustered at the bank and LBO firm level. *t*-statistics are reported in parentheses. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

Table 4
Transaction fees

Panel A: Spread						
	(1) OLS Coeff.	(2) OLS Coeff.	(3) OLS Coeff.	(4) OLS Coeff.	(5) 2OLS Coeff.	(6) Spread net of fees
Previous fee transactions (same bank)	−3.64 *** (3.31)	−3.57 *** (3.21)	−2.93 ** (2.57)	−3.15 *** (2.81)	−2.59 ** (2.13)	−2.44 (1.12)
Previous fee transactions (other banks)	−1.32 (1.47)	−1.20 (1.19)	−0.14 (0.13)	−1.48 (1.45)	−1.86 * (1.82)	2.98 (1.62)
Bank relationship	—	—	—	−23.52 ** (2.37)	−54.47 * (1.96)	−22.26 (1.18)
Previous loans	—	−0.75 (0.33)	—	−0.81 (0.36)	−0.89 (0.40)	3.83 (0.68)
Fund size	—	—	−9.17 ** (2.29)	—	—	—
Benchmark controls	yes	yes	yes	yes	yes	yes
Observations	1,590	1,590	1,518	1,590	1,590	381
Adj. R-squared	0.36	0.36	0.37	0.36	0.36	0.55

Panel B: Maximum debt to EBITDA						
	(1) OLS Coeff.	(2) OLS Coeff.	(3) OLS Coeff.	(4) OLS Coeff.	(5) 2OLS Coeff.	
Previous fee transactions (same bank)	0.08 *** (3.08)	0.06 ** (2.38)	0.06 ** (2.31)	0.06 ** (2.20)	0.05 (1.92)	*
Previous fee transactions (other banks)	−0.01 (0.56)	−0.04 (1.51)	−0.04 (1.35)	−0.04 (1.43)	−0.03 (1.39)	
Bank relationship	—	—	—	0.68 (2.57)	** (2.20)	**
Previous loans	—	0.18 (3.00)	—	0.18 (2.92)	*** (2.83)	***
Fund size	—	—	0.18 (1.57)	—	—	
Benchmark controls	yes	yes	yes	yes	yes	
Observations	474	474	448	474	474	
Adj. R-squared	0.53	0.54	0.55	0.54	0.51	

Spread is measured in basis points and corresponds to the total cost (interest rate and fees) paid over LIBOR for each dollar drawn down under the loan commitment. *Spread net of fees* excludes the upfront fees paid to the lead bank. The dependent variable in Panel B is the *Maximum total debt to EBITDA* covenant set at loan origination. *Previous fee transactions* is calculated as the logarithm of one plus the sum of transactions for the lead bank (*same bank*) or for all advisors other than the lead bank (*other banks*). Each regression includes all of the benchmark specification controls (*Deal amount*, *Maturity*, *Unsecured*, *Number of lenders*, *Lender rank*, *Traded loan*, *New lead bank-borrower*, *Book leverage*, *EBIT to assets*, *Sales to assets*, *Working capital to assets*, *Retained earnings to assets*, *No Compustat data*, and *No loan rating*) and year, industry, and credit rating fixed effects. Regression (5) is the second stage of the 2SLS specification where *Bank relationship (amount)* is instrumented with the difference in lagged relationship and the lagged relationship of the PE firm's second-largest bank. See the appendix for the first stage of the 2SLS specification. Standard errors are clustered at the bank and LBO firm level. *t*-statistics are reported in parentheses. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

H1 and H2 appear to be additive. When *Bank relationship* is included in the specifications, the coefficients on *Bank relationship* and cross-selling (*Previous fee transactions (same bank)*) remain of the same sign (Table 4, Panels A and B, specification 4). While the level of statistical significance is reduced,

we continue to estimate a negative association between *Bank relationship* and loan spread, and a positive association between *Bank relationship* and *Maximum debt to EBITDA*. A one-standard-deviation increase in both *Bank relationship* and *Previous fee transactions (same bank)* results in a 17 basis point lower spread and a 0.4x higher maximum debt to EBITDA covenant.

These results are consistent with reduced information asymmetry and with cross-selling affecting loan terms. However, one should be cautious in strictly interpreting the relative magnitude of the two effects. As the variable for cross-selling potential is noisy and often equal to zero, the bank relationship variable could be picking up both reduced information asymmetry and cross-selling potential.

We also attempt to distinguish between cross-selling and information in the final specification of Panel A, where the dependent variable is the spread net of fees rather than the all-in-spread. While this measure does not capture the cost of the loan for the borrower, it allows us to test H3. The spread net of fees is earned by all syndicate members, and it should be lower when information asymmetry is reduced (H1). In contrast, cross-selling benefits (H2) accrue only to the lead bank. Therefore, syndicate members should not be willing to accept lower spreads. Consistent with our hypothesis, when information asymmetry inefficiencies are lowered by the repeated relationship, the loan spread shared by the lending syndicate is lower. The coefficient on *Previous fee transactions* declines more than the coefficient on *Bank relationship*, but neither are statistically significant, perhaps as a result of the lower number of observations.

In unreported analyses, we examined the loan syndicate structure as a function of the bank relationship and previous fee transactions. We expected the information benefits of the banking relationship to allow for broader syndication. While the direction of the effect was as predicted, it was not statistically significant.

Note that concerns about the conflating of cross-selling and bank relationship are reduced in the instrumental variable approach. In the first-stage specification, the addition of the cross-selling measure does not affect the relationship between the instrument and the bank relationship, even though cross-selling is significantly associated with the cross-selling measure. The instrumented bank-relationship variable should be unrelated to cross-selling prospects after controlling for cross-selling. It is also worth noting that in the presence of cross-selling and the absence of costs of information asymmetry, LBO firms would never need to switch banks because the banks cannot hold up the LBO firms.

3.2 Economic impact: Equity returns

To calculate the direct economic impact of LBO firms' bank relationships, we must make assumptions about the composition and horizon of LBO firms' investment portfolios. Based on the coefficients estimated in Table 4, specification 4, lower spreads result in annual savings of approximately \$0.55 million

for each investment made by an LBO firm that has both one standard deviation stronger bank relationships (a 0.287 increase in *Bank relationship*) and one standard deviation stronger cross-selling prospects (an increase of 3.31 in log fees). This assumes that the PE firm pays a spread that is 17.19 basis points lower ($-23.52 \times 0.287 + -3.15 \times 3.31$) on a loan of mean size (\$321 million). The present value of the lower interest rate, received over four years and discounted at 8% (LIBOR of 5% + mean loan spread of 300 basis points), is \$1.83 million, which yields an incremental return of 85 basis points for an initial equity contribution of \$214 million (assuming that equity is 30% of capital structure and loans are 44% of capital structure, as estimated by S&P aggregate data).³⁰

Similarly, if the LBO firm takes advantage of the higher maximum debt to EBITDA covenant resulting from its one standard deviation stronger bank relationships and its one standard deviation stronger cross-selling prospects to replace equity with debt, equity returns will be 3.50 percentage points higher. The covenant impact is calculated from the difference between the return to equity on an investment of \$1,000 million financed with \$700 million in total debt (and an implied \$129 million in EBITDA assuming the mean debt to EBITDA covenant) vs. \$750 million in total debt (an increase in total debt to EBITDA from the mean of 5.41x to 5.80x). The 0.38x increase in the debt to EBITDA covenant is from $(0.68 \times 0.287 + 0.06 \times 3.31)$. Of course, the latter marginal effect is associated with higher risk.

The instrumented coefficients suggest that the effect is even larger. Overall, the potential increases in equity returns range from 4% to 9% (on a non-risk-adjusted basis). This is economically significant in the context of mean equity returns of 17%. The 0.85% (1.20% using instrumental variables estimates) increase in equity return due to lower spread is risk independent and large in comparison to performance estimations (alpha) that range from 2% to negative for leveraged buyouts in general (for example, see [Groh and Gottschalg 2007](#); [Kaplan and Schoar 2005](#); and [Phalippou and Zollo 2006](#)). The actual contribution of LBO firms' bank relationships to equity returns is likely to be lower because the hypothetical increase in equity returns assumes that none of the benefits of better bank loan terms are passed on to companies that the LBO firms buy via higher valuations. The results reported below suggest that some of these benefits may actually be passed on to sellers.

Another context for evaluating the economic impact of loan terms is found in [Kaplan's \(1989\)](#), and [Guo, Hotchkiss, and Song's \(2010\)](#) estimates of operating

³⁰ Equity contribution and exit horizon are based on annual medians for LBOs of public companies as compiled by [Guo, Hotchkiss, and Song \(2010\)](#). Bank financing percentage is based on S&P data including all LBOs with syndicated bank loans. The EBITDA calculation assumes a total enterprise value growth rate of 6% and debt payoff of 25%, which were quoted to us by practitioners and correspond to industry averages. The discount rate is based on the median loan price from DealScan. Use of this discount rate implicitly assumes that the bank debt is not risky. Arguably, the discount rate should reflect the probability that the company goes bankrupt and is unable to achieve these savings over the life of the loan. Each 100-basis-point increase in the discount rate assumption reduces the IRR impact by approximately 2 basis points.

changes in LBO companies. [Kaplan \(1989\)](#) estimates an industry-adjusted change in net cash flow to sales over four years of 28.3% (based on 16 pre-1989 buyouts). The equity internal rate of return (IRR) impact implied by this four-year growth rate is at least 6.4%, assuming that sales and the valuation multiple for cash flow remain unchanged. [Guo, Hotchkiss, and Song \(2010\)](#) use more recent data on 33 buyouts to estimate a four-year mean industry-adjusted change in cash flow to sales of 9.0%, which has an implied IRR impact of at least 2%.

We also directly examine the impact of loan terms on equity returns by looking at a subsample of investments for which we have equity returns data obtained from marketing materials for 30 LBO firms.³¹ As the previous calculations suggest, the terms that LBO firms receive on their bank loans are positively associated with equity returns. Even after controlling for LBO firm fixed effects, investments with higher maximum debt to EBITDA covenants have significantly higher equity returns in the sample of 82 deals for which we have equity returns data. Investments with lower spreads also have higher equity returns, although the effect is not statistically significant.

The aforementioned calculations address only the direct impact of different loan terms on the LBO firm. However, this is likely to be the lower bound of the actual effect, as we are unable to calculate the economic impact of the availability or the speed with which deal financing can be raised, both of which are likely to be competitive advantages for LBO firms.

More broadly, the economic magnitude of our findings is in line with several studies on relationship lending. For example, [Bharath, Dahiya, Saunders, and Srinivasan \(2008\)](#) use a similar sample of loans to document that strong banking relationships are associated with interest rates that are 5 to 15 basis points lower. However, contrary to our results, their findings hold primarily for smaller firms in the sample.

3.3 Economic impact: Transaction prices

In light of our findings on loan terms, we would like to analyze who benefits from those terms. [Kaplan and Stein \(1993\)](#) find that LBO transaction prices are responsive to lending market conditions. Their findings imply that LBO sponsors pass on better loan terms to sellers in the form of higher transaction prices.

We matched 732 of our transactions to the Capital IQ database. However, only limited information is available for those transactions. Of the 732 matches, we find information on total enterprise value for 468 and details on total enterprise value to EBITDA for only 112. In addition, for companies that were previously publicly traded, we find information on the premium paid over

³¹ The sample of marketing memoranda was obtained confidentially from an anonymous limited partner and is described in more detail in [Kovner \(2009\)](#). It includes actual equity returns, gross of fees, for all of the LBO firms' investments. Details are available from the authors.

Table 5
Impact of loan terms on transaction price

	Premium 1 month prior		Enterprise value to EBITDA		Enterprise value	
	Coeff.	Coeff.	Coeff.	Coeff.	Coeff.	Coeff.
Bank relationship	7.85 (0.47)	—	2.42 (1.25)	—	708.32 * (1.75)	—
Previous loans	−3.50 (1.53)	−5.09 * (1.88)	−0.44 ** (2.06)	−0.47 * (1.95)	−32.77 (1.01)	−47.75 (1.36)
Previous fee transactions (same bank)	—	−0.73 (0.26)	—	0.06 (0.43)	—	39.72 ** (2.08)
Previous fee transactions (other bank)	—	2.63 (0.96)	—	0.03 (0.16)	—	9.81 (0.78)
Observations	66	66	114	114	475	475
Adj. R-squared	0.76	0.79	0.64	0.64	0.55	0.56

The dependent variable in the first two specifications is *Premium 1 month prior*, which is calculated as the percentage change in the stock purchase price at the time of the transactions relative to the stock price one month earlier. The dependent variable in specifications 3 and 4 is *Enterprise value to EBITDA*, which is the total enterprise value of the transaction divided by the company's EBITDA for the last twelve months. The dependent variable in specifications 5 and 6 is *Enterprise value*, which is the total enterprise value of the transaction. Each regression includes the following benchmark specification controls: *Deal amount*, *EBIT to assets*, *Sales to assets*, *Working capital to assets*, *Retained earnings to assets*, and *No Compustat data*, as well as year and industry fixed effects. The absolute value of the *t*-statistic is in parentheses. Standard errors are clustered at the PE firm level. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

the one-month-prior stock price for 66 transactions. We then winsorize the sample at the 5% level to correct for outliers, which further reduces the number of observations. The results of a regression of valuation metrics on our variables of interest are presented in Table 5.

While the results are mostly not statistically significant, *Bank relationship* is positively related to the price paid. The results are similar for cross-selling, with a significant positive coefficient in the enterprise value specification but non-significant coefficients for the other valuation metrics. The estimated coefficient on *Bank relationship* of 2.42 is actually higher than that estimated in the total debt to EBITDA specifications, suggesting that all of the benefit of better loan terms accrues to sellers. This implies that the value of bank relationships for LBO firms does not lie in increased equity returns but in the fact that such relationships allow them to bid aggressively. However, this result must be interpreted cautiously given the limited number of observations.

3.4 Robustness: Fixed effects

It is possible that there is unobservable heterogeneity across LBO firms and their investments that is correlated with bank relationships, previous transactions, and loan terms. For example, if some LBO firms are consistently of better quality, then every bank would want to continue working with them. Better LBO firms would have high bank relationships.

Perhaps, instead, the bank determines both the level of relationship and the loan terms. For example, larger banks are more likely to be high-relationship

Table 6
Robustness: Fixed effects

Dependent variable:	Spread			Maximum debt to EBITDA		
	Benchmark: Table 4, Panel A (4)			Benchmark: Table 4, Panel B (4)		
	Coeff.	Coeff.	Coeff.	Coeff.	Coeff.	Coeff.
Bank relationship	-26.19*** (10.97)	-25.16** (2.42)	-20.73* (1.78)	0.40 (0.29)	0.70*** (2.69)	0.55 (1.63)
Previous fee transactions (same bank)	-2.76 (1.18)	-3.98*** (2.97)	-4.22** (2.35)	0.05 (0.03)	0.06* (1.93)	0.06 (1.44)
Previous fee transactions (other banks)	-1.54 (1.35)	-1.34 (1.33)	-1.84 (1.36)	-0.06 (0.03)	-0.03 (1.22)	-0.08** (2.08)
Previous loans	-2.60*** (3.53)	-0.02 (0.01)	-1.42 (0.38)	0.20 (0.13)	0.17** (2.05)	0.18 (1.21)
Controls:						
Benchmark controls	yes	yes	yes	yes	yes	yes
LBO firm fixed effects	yes	no	yes	yes	no	yes
Bank fixed effects	no	yes	yes	no	yes	yes
Observations	1,590	1,590	1,590	474	474	474
Adj. R-squared	0.43	0.42	0.48	0.64	0.61	0.69

Spread is measured in basis points and corresponds to the total cost (interest rate and fees) paid over LIBOR for each dollar drawn down under the loan commitment. *Maximum debt to EBITDA* is the covenant level at loan origination. Independent variables are in logs, except for dummy variables. Each regression includes all of the benchmark specification controls (*Deal amount*, *Maturity*, *Unsecured*, *Number of lenders*, *Lender rank*, *Traded loan*, *New lead bank-borrower*, *Book leverage*, *EBIT to assets*, *Sales to assets*, *Working capital to assets*, *Retained earnings to assets*, *No Compustat data*, and *No loan rating*) and year, industry, and credit rating fixed effects. Standard errors are clustered at the bank level and at the borrower level for specifications with LBO firm fixed effects, and at the LBO firm level for specifications with bank fixed effects alone. *t*-statistics are reported in parentheses. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

banks if every loan is allocated according to market share. Furthermore, larger banks might have lower monitoring costs or punish firms more severely. Loans underwritten by larger banks would then be on more favorable terms, and larger banks would also have high relationships with most LBO firms. To address this possibility, we reexamine our central results by introducing LBO firm and bank fixed effects. Overall, our findings are robust to the inclusion of controls for LBO firm and bank fixed effects.

Table 6 shows the results when fixed effects for each LBO firm are added to our primary specification (Table 4, Panels A and B, specification 4). Coefficients on *Bank relationship* remain negative for the spread regression and are significant at the 1%, 5%, or 10% level. Coefficients for the *Maximum debt to EBITDA* specification remain positive but are not statistically significant for *Bank relationship* in all specifications. This probably reflects the reduced estimation power arising from the relatively limited within-group variation.

To better understand the LBO firm fixed effects estimation results presented in Table 6, consider the example of an LBO firm, KKR, and a lead bank, Bank of America. Controlling for LBO firm fixed effects means that we estimate how KKR’s loan terms on loans made to its LBOs by Bank of America (underwriter of 30% of KKR’s deals on average) differ from KKR’s loan terms on loans

made to its LBOs by Credit Suisse (underwriter of 12% of KKR's past deals, on average). The table indicates that, all else equal, KKR should expect 4.7 basis points lower spreads from Bank of America than from Credit Suisse, a finding that supports H1.

While they remain of the same sign, the coefficients on our variables of interest in the LBO firm fixed effects specifications are reduced in size in some cases. This makes sense, particularly in the cross-selling case, because opportunities for future business may be fixed at the LBO firm level. Firms with more capital pay more fees. It is possible that different PE firms are more likely to engage in transactions that generate banking fees in ways other than size differences. For example, KKR's average holding period is longer than Blackstone's. Thus, for the same amount of equity invested, the expected fees would be lower for KKR than they are for Blackstone. If this effect is fixed at the firm level, LBO firm fixed effects may actually explain some of the variation in which we are interested.³²

The second and fifth specifications of Table 6 show the results when fixed effects for each lead bank are added. There is less variation within banks for our key variables, which makes it more difficult to identify the effects. The average of the standard deviation of *Bank relationship* within each bank is only 0.19, compared to 0.29 for the sample. However, our main results are robust to bank fixed effects. For example, Bank of America's loan terms to KKR when Bank of America has led 30% of KKR's previous deals on average will not be as good as their loan terms to Blackstone, as Bank of America has led 38% of Blackstone's past deals on average. We continue to find evidence that bank relationships and prospects for fees improve loan terms, even when considering only variation within lead banks.

We also estimate bank and LBO firm fixed effects together and find consistent results. This estimation allows for specific LBO firms to receive and for specific banks to grant consistently different loan terms. Given the relatively limited nature of our sample, it is not surprising that while the signs remain consistent, the *t*-statistics are reduced and statistical significance disappears for some of the coefficients on *Previous fee transactions (same bank)* in the *Maximum debt to EBITDA* estimation. While we include bank, LBO firm, and bank and LBO firm fixed effects, we unfortunately do not have enough observations in our sample to allow for bank x LBO firm fixed effects.

As these specifications are attempts to directly understand fixed differences in bank and LBO firms' relationships, we do not include the instrumental variable estimates in these models. Instead, we attempt to directly address concerns that bank or LBO firm characteristics are related to both bank

³² In unreported regressions, we also control for fixed effects by firm and time period, allowing the fixed effects for firms to vary with each time period. Time periods are assigned by fund, with the assumption that firm characteristics are relatively stable within funds, if not within firms. The results are qualitatively similar.

relationships and loan terms. In unreported regressions, we consider several mechanisms that might influence the endogenous matching process between the LBO firm and the bank, and thereby affect both the bank relationship and loan pricing. Specifically, we include the following controls: (i) the bank's industry expertise; (ii) whether the bank was an investor in the LBO fund; (iii) the LBO firm's age; and (iv) the LBO firm's industry expertise. Our results are robust to the inclusion of these controls, although the fixed effects specifications would capture these results, assuming that these characteristics are fixed over time.³³

In some cases, the loans we observe were arranged as "stapled" financing. In stapled financing, the investment bank that is retained to sell a company also provides financing to potential buyers. In this case, the bank assigns a team to work with each potential bidder and the terms of the loans offered to each bidder may not necessarily be the same. An LBO firm's bank relationship might be less important in this setting, and we would be less likely to find a relationship between our variables of interest and bank loan terms. In unreported estimations, we exclude financings in which the lead bank also served as the target company's financial advisor (likely stapled financings) and verify that bias in the result is economically insignificant.

3.5 Robustness: Ex-post performance

In our final set of robustness tests, we look at future performance as a proxy for the borrower's (portfolio company) unobservable expected performance. If our measures are correlated with the unobservable quality of the current loan, we would expect them to be correlated with ex-post loan performance. We examine the following measures of ex-post loan performance: (i) bankruptcy; (ii) presence and time to renegotiation of the loan covenants; and (iii) loan performance on the secondary market. Table 7 presents the findings related to these measures. Overall, the analysis reveals no ex-post evidence that our bank relationship measures are correlated with ex-ante unobservable quality.

Bankruptcy can result in missed coupon payments or a failure to recover the full loan value, and thus it serves as an ex-post measure of bank loan quality. To identify bankruptcies, we use information on the outcomes of all of the investments of 30 of the LBO firms in our sample (see Kovner 2010 for a detailed description of this sample) and information from the Bankruptcy Insider database. Bankruptcy Insider is a product offered by Daily Deal that is targeted at bankruptcy professionals and offers consolidated news and

³³ In order to confirm that no single LBO firm was driving the results, we also estimate our specifications after dropping one LBO firm at a time. This produces similar results. We also examine the effects after weighting observations by the amount of the syndicated loan, and our results are unchanged. These results are available from the authors upon request.

Table 7
Robustness: Ex-post Performance Measures

Performance measure		Bank relationship	Previous fee transactions (same bank)	Previous fee transactions (other banks)	Obs.	Adj. R ²
Bankrupt	Coeff.	0.0308 (1.28)	−0.0041*** (2.67)	−0.0004 (0.18)	1,246	0.06
Renegotiation	Coeff.	0.0437 (1.54)	0.0136*** (3.44)	0.0021 (0.74)	1,590	0.12
Days to renegotiation	Coeff.	55.0373 (1.12)	−17.5199*** (3.37)	−0.9652 (0.18)	1,590	0.37
Adj. loan return, 12 months	Coeff.	0.0197 (0.23)	0.0037 (0.47)	−0.0038 (0.44)	262	0.28
Adj. loan return, 36 months	Coeff.	−0.0167 (0.36)	0.0071 (1.20)	−0.0053 (1.08)	262	0.17

The dependent variable is the performance measure. *Bankrupt* is a dummy variable equal to 1 if the company went bankrupt after the loan origination and 0 otherwise. *Renegotiation* is a dummy variable equal to 1 if the loan was renegotiated and 0 otherwise. *Days to renegotiation* is the number of days that passed before the loan was renegotiated or the maturity of the loan. *Adj. loan return* is the post-origination loan return adjusted for leveraged loan market returns in the same time period. All specifications include controls for two-digit SIC industry and year fixed effects. Standard errors are clustered at the bank and LBO firm level. *t*-statistics are reported in parentheses. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

information about bankruptcies. Of the 1,244 investments for which we have outcome³⁴ and DealScan information, 55 (4%) ended in bankruptcy.³⁵ Although *Bank relationship* does not appear to be associated with higher-quality borrowers (a lower probability of bankruptcy), bankruptcy is a relatively infrequent outcome and a borrower may be troubled but never file for bankruptcy.

We also measure ex-post borrower quality using the loan’s renegotiations (if any). Borrowers are likely to renegotiate when they have underperformed financially.³⁶ We also consider time to renegotiation because loans that are renegotiated sooner are more likely to be those encountering trouble. In this regard, being on better terms with a lead bank might make a firm more likely to engage in a renegotiation for non-negative reasons. However, bank amendments typically result in the payment of bank fees, which should mean that borrowers attempt to avoid them whenever possible. Our relationship measure does not appear to be related to ex-post borrower quality as measured in loan renegotiation terms. On the other hand, the metric of cross-selling

³⁴ The remaining companies have names that were not found in the Bankruptcy Insider database or were difficult to confirm (such as “Security Co. Inc.”).

³⁵ While we measure bankruptcy with a binary variable, we estimate a linear regression, rather than a logit or probit model, due to the substantial number of binary fixed effects.

³⁶ We exclude amendments that do not result in any significant changes to loan terms. However, it is possible that some of the included amendments are not necessarily a signal of financial underperformance, as they may be related to acquisitions (for example, in a roll-up strategy).

opportunities is associated with more-troubled borrowers (positively associated with the likelihood of renegotiation and negatively associated with days to renegotiation), which should create a bias against our results.

Finally, we look at the aftermarket loan performance over two time horizons, which are adjusted to reflect the performance of the leveraged loan market (based on Thomson Reuters Secondary Market intelligence (SMi) database of secondary loan market prices) over 12 months and 36 months. In neither case are our variables of interest significantly related to aftermarket loan performance. The lack of a relationship between *Bank relationship* and loan performance may simply reflect the fact that these loans are appropriately priced by the market. Unfortunately, the loan aftermarket data begin in 1998 and are therefore available for only a subset of the data, so that the sample is limited to fewer than 300 observations.

4. Conclusions

An understanding of LBO firms is important both because of the scale of their operations and in light of their role as a disciplinary force in public markets. While private equity generally appears to create economic value, the specific role of LBO firms as financial intermediaries has not been explored. By presenting evidence that bank relationships lead to different financing terms for LBO firms, we show that LBO firms have an investment or management skill that is not easily communicated or transferred, and that LBO firms play a role as intermediaries in accessing debt markets.

LBO firms' repeated interactions with banks create opportunities to reduce the costs associated with information asymmetries. Prospects for cross-selling to LBO firms' investment portfolios also appear to improve loan terms. An LBO firm that has a one standard deviation higher bank relationship and a one standard deviation higher cross-selling potential will have a spread that is 17.2 basis points lower (5%) and a maximum debt to EBITDA covenant that is 0.38x higher (7%), which gives a potential increase in equity returns of 4% to 9% (on a non-risk adjusted basis). These results suggest a new way in which LBO firms enhance value—through their repeated interactions with banks.

It is important to understand the implications of our findings for non-sponsored borrowers. The best method to assess the importance of LBO firms' banking relationships would be to compare the terms of a sample of leveraged loans to a set of comparable loans to companies without financial sponsors. However, there are almost no truly comparable loans that do not have financial sponsors. Given the lack of comparable non-sponsored transactions, we propose a loan-market-based explanation—it is possible that lenders are unwilling to finance such transactions in the absence of repeated relationships.

Appendix

First-stage Regression: Instrumented Bank Relationship

	(1) Table 2 Coeff.	(2) Table 3 Coeff.	(3) Table 4A Coeff.	(4) Table 4B Coeff.
Difference lagged bank relationship	-0.24*** (7.89)	-0.21*** (2.84)	-0.23*** (7.53)	-0.21*** (2.80)
Lagged bank relationship (2nd)	0.51*** (5.80)	0.24 (1.03)	0.48*** (5.34)	0.24 (1.04)
Previous loans	0.02*** (3.36)	0.03* (1.91)	0.02*** (3.31)	0.02 (1.55)
Previous fee transactions (same bank)	—	—	0.02*** (6.33)	0.01 (1.31)
Previous fee transactions (other bank)	—	—	-0.01*** (4.23)	0.00 (0.57)
Unsecured	-0.03 (0.90)	0.06 (0.86)	-0.04 (1.07)	0.06 (0.82)
Number of lenders	0.00 (0.71)	0.00 (0.08)	0.00 (0.68)	0.00 (0.05)
Deal amount	0.03*** (2.64)	0.02 (1.10)	0.02* (1.71)	0.02 (0.92)
Maturity	0.02 (0.74)	0.07 (1.61)	0.00** (2.24)	0.00 (0.81)
Book leverage	-0.07 (0.46)	-0.07 (0.86)	0.01 (0.52)	0.07 (1.58)
EBIT to assets	-0.02 (1.24)	-0.03 (1.24)	0.01 (0.05)	-0.03 (0.63)
Sales to assets	0.09* (1.88)	0.14 (0.96)	-0.02 (1.05)	-0.02 (1.07)
Working capital to assets	-0.02 (1.15)	0.06 (1.14)	0.08* (1.68)	0.13 (0.87)
Retained earning to assets	-0.01 (0.48)	0.01 (0.31)	-0.03 (1.62)	0.06 (1.07)
No Compustat data	-0.01 (0.85)	0.00 (0.07)	-0.01 (0.57)	0.01 (0.34)
New lend bank - borrower	0.04 (0.95)	0.06 (0.60)	-0.01 (0.54)	0.00 (0.34)
No loan rating	-0.17** (2.15)	0.00 (0.00)	0.07 (1.45)	0.07 (0.72)
Traded loan	0.12** (2.13)	0.10 (0.91)	0.02 (1.03)	0.06* (2.01)
Lender rank	0.02 (0.23)	0.08 (0.36)	0.00*** (7.72)	0.00*** (2.97)
Fixed effects:				
Year	yes	yes	yes	yes
Industry (2 digit SIC code)	yes	yes	yes	yes
Credit rating	yes	yes	yes	yes
Observation	1,590	474	1,590	474
Adj. R-squared	0.17	0.18	0.19	0.18

The specifications are the first stage of the specifications for the instrument bank-relationship variable shown in Tables 2–4. The dependent variable is *Bank relationship (amount)*, which is the dollar value of loans in the previous five years underwritten by the lead bank divided by the total dollar value of all loans sponsored by the PE firm. The instruments are *Difference in lagged bank relationship*, which is the difference between the highest and second-highest bank relationships of the PE firm before the current deal, and *Lagged bank relationship* (2^{nd}), which is the level of the second-highest bank relationship of the PE firm before the current deal. The remaining variables are those used in the second-stage estimations, including *Previous loans* (the log of one plus the dollar value of loans associated with transactions sponsored by the LBO firm in the previous five years). Specifications (3) and (4) also include *Previous fee transactions*, which is the logarithm of one plus the sum of transactions for the lead bank (*same bank*) or for all advisors other than the lead bank (*other banks*). Each regression includes all of the benchmark specification controls (*Deal amount*, *Maturity*, *Unsecured*, *Number of lenders*, *Lender rank*, *Traded loan*, *New lead bank-borrower*, *Book leverage*, *EBIT to assets*, *Sales to assets*, *Working capital to assets*, *Retained earnings to assets*, *No Compustat data*, and *No loan rating*) and year, industry, and credit rating fixed effects. Standard errors are clustered at the bank and LBO firm levels. *t*-statistics are reported in parentheses. ***, **, and * indicate *p*-values of 1%, 5%, and 10%, respectively.

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